Remaining Useful Life of Turbofan Engines

Phase 1: engineered features and classical ML on NASA C-MAPSS FD001

cmapss-rul-hybrid team

Phase 1 review

The problem

- Predict **remaining useful life (RUL)**: how many operating cycles an engine has left before failure, from its sensor history.
- Maintenance is scheduled on this number: predicting too much life risks a failure in service, too little wastes engine life.
- Project question: do **hand-engineered degradation features** with classical regressors (Phase 1) or **sequence models** that learn from raw sensor windows (Phase 2) predict RUL better?

Phase 1 deliverable

A leakage-free baseline on FD001: Random Forest, XGBoost and Elastic Net, each trained on raw inputs and on engineered features, scored on the official test engines.

Data: NASA C-MAPSS

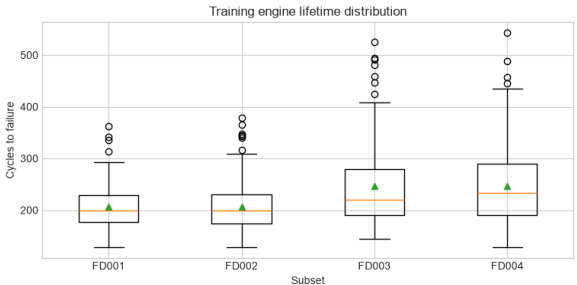
Subset	Cond.	Faults	Train	Test
FD001	1	1	100	100
FD002	6	1	260	259
FD003	1	2	100	100
FD004	6	2	249	248

Engines per split. Simulated data, Saxena et al. (2008).

- One row per engine cycle: `unit`, `cycle`, 3 operating settings, 21 sensors.
- Training engines run to failure; test engines stop early.
- Training target: final cycle − current cycle.
- Test: scored once per engine, at its **last cycle**, against `RUL_FD001.txt`.

Takeaway: FD001 (one operating condition, one fault) is the Phase 1 subset; 20,631 training rows and 13,096 test rows.

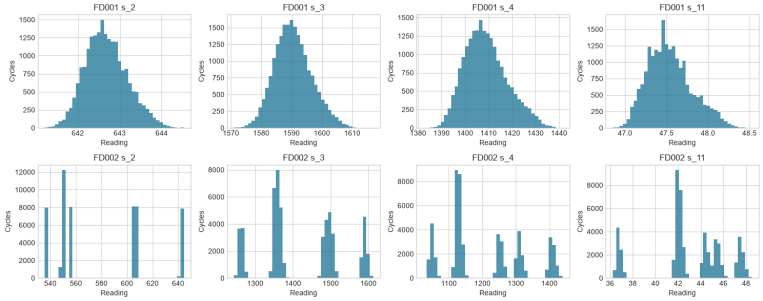
EDA 1: lifetimes vary widely, so split by engine



- FD001 lifetimes: 128 to 362 cycles, median 199.
- Rows of one engine are strongly related, so a row-level split would leak an engine into both train and validation.
- The long right tail makes the uncapped target span a wide range, which inflates early-life errors.

Takeaway: All splits are by engine (`GroupKFold` on `unit`).

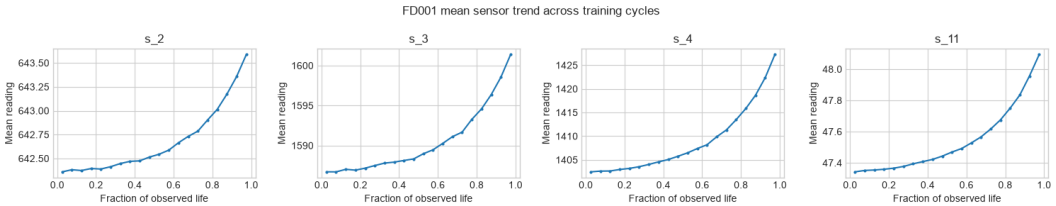
EDA 2: operating conditions dominate raw sensor levels



FD001 sensors are unimodal; in FD002 the same sensors split into one mode per operating condition, a shift much larger than degradation. Six FD001 sensors ($s_1, s_5, s_{10}, s_{16}, s_{18}, s_{19}$) are exactly constant.

Takeaway: Normalize each sensor within its operating regime, and drop constant sensors, fitted on training folds only.

EDA 3: degradation is slow, then accelerates



Mean FD001 reading against fraction of observed life. Sensors barely move for the first half of life, then rise increasingly steeply.

- Level *and* recent slope carry information $\Rightarrow$ rolling mean and rolling slope features.
- Early cycles look alike while their true RUL differs widely $\Rightarrow$ early-life errors will be large under an uncapped target.

Evaluation protocol

Guardrails (from CONTRIBUTING)

- Split by engine only, never by row.
- Sensor selection, regime clusters and scaling refitted *inside* each training fold.
- Rolling windows never cross engine boundaries or look ahead.
- Uncapped RUL is the primary target; capping is an ablation.

Tuning and test

- Grid search, 3 engine-grouped folds over 100 training engines, seed 42, selected by CV RMSE.
- Best candidate refitted on all 100 training engines.
- 100 test engines scored once at their last cycle.
- Metrics: RMSE, MAE, NASA score, and RMSE by true-RUL stage: late $\leq 50 <$ mid $\leq 100 <$ early.

Feature engineering: 155 predictors per cycle

1. **Selected sensors:** drop exactly constant channels $\Rightarrow$ 15 of 21 kept.
2. **Regime z-score:** $z_{u,t,s} = (x_{u,t,s} - \mu_{r,s})/\sigma_{r,s}$, with K-means regimes on the operating settings.
3. **Rolling features** over trailing windows $w \in \{5, 10, 20\}$:
   - mean m (local level)
   - least-squares slope b (rate of change)
   - health change $m - z_{t_0}$ (vs. engine's first cycle)
4. **Context:** `cycle`, 3 settings, regime.

Predictor count

$$5 + S(1 + 3W) = 5 + 15 \cdot (1 + 3 \cdot 3) = 155$$

Raw reference: `cycle`, 3 settings and all 21 sensors.

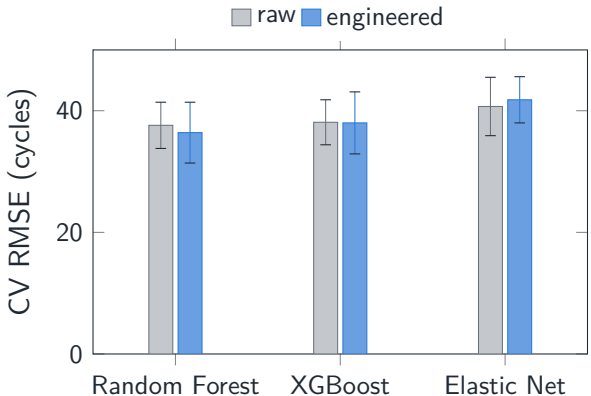
All features are **causal**: they use only the current and past cycles of the same engine.

Models

Model	Why it is here	Grid searched
Random Forest	Averages deep trees to cut variance; non-linear, no scaling needed, predictions stay in the training RUL range.	trees $\{50, 100\}$, min leaf $\{1, 3\}$, `max_features`=0.5
XGBoost	Boosts shallow trees to cut bias; usually the strongest tabular model.	trees $\{100, 200\}$, depth $\{3, 5\}$, learning rate fixed at 0.3
Elastic Net	Linear reference with $\ell_1 + \ell_2$ penalty; handles the many correlated rolling features; interpretable.	$\alpha\{0.1, 1.0\}$, ℓ_1 ratio $\{0.2, 0.8\}$, standardized inputs

All three minimize squared error, so they estimate the conditional mean RUL. The asymmetric NASA score is reported but not optimized.

Cross-validation does not separate the models



Mean ± standard deviation over 3 engine-grouped folds.

- Every gap between models or feature sets (at most 5.4 cycles) is about the size of one fold's standard deviation (3.7 to 5.1).
- With only 3 folds, the ranking could change with a different seed.
- Several selected settings sit on the edge of their grid.

Takeaway: CV alone does not establish a preferred model or feature set.

Official FD001 test results (uncapped RUL)

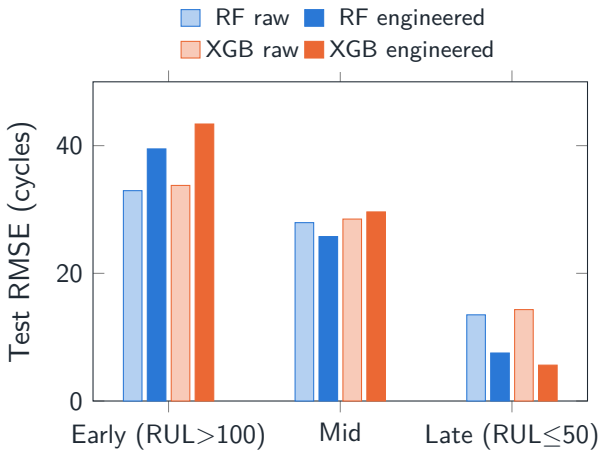
	Engineered features			Raw inputs		
Model	RMSE	MAE	NASA	RMSE	MAE	NASA
Random Forest	27.55	**17.67**	145,410	**26.15**	19.22	8,414
XGBoost	30.50	19.82	61,202	26.84	19.19	7,002
Elastic Net	28.39	23.57	7,777	31.08	25.80	**6,903**

100 test engines, each scored at its last recorded cycle. Lower is better. Bold = best in column group.

- **No single winner:** raw Random Forest has the lowest RMSE, engineered Random Forest the lowest MAE, raw Elastic Net the lowest NASA score.
- Engineered features lower RMSE only for Elastic Net (31.08 $\rightarrow$ 28.39); for both tree models they raise it.

Caveat: One test set of 100 engines and no confidence intervals: a difference of 1 to 2 RMSE may not be real.

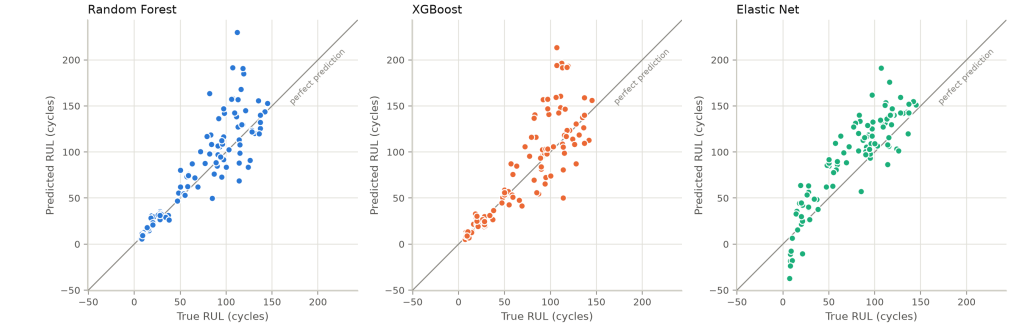
Where the errors are: by true-RUL stage



33 early, 34 mid and 33 late engines.

- **Near failure, engineered features win clearly:** RF 13.5 $\rightarrow$ 7.5, XGB 14.3 $\rightarrow$ 5.6. This is where maintenance is scheduled.
- **On young engines they lose:** RF 33.0 $\rightarrow$ 39.5, XGB 33.8 $\rightarrow$ 43.4.
- Elastic Net stays poor late (24.8): a linear model cannot follow the sharp end-of-life drop.

Predicted vs. true RUL (engineered models)



- Tight near zero RUL, wide scatter above about 80 cycles.
- **NASA score is driven by single engines:** test engine 1 (31 cycles seen, 112 left) is predicted at 230 by engineered RF, which is 92% of its 145,410 score. Read NASA together with RMSE and MAE.
- Elastic Net predicts negative RUL for 7 engines (not clipped).

The headline depends on the target definition

Why young engines are hard. With an uncapped target, an engine at cycle 31 that still looks new could have 100 or 300 cycles left; the sensors cannot tell. The models fall back on age: Tree SHAP attributes far more of each engineered prediction to `cycle` than to any sensor.

Preliminary check: Random Forest trained with RUL capped at 125

Training target	Engineered	Raw
Uncapped (primary)	27.55	**26.15**
Capped at 125	**16.4**	18.7

Test RMSE against the true (uncapped) test RUL. Random Forest only, one run; not yet in `results/metrics/`.

Caveat: Under a capped target the raw-vs-engineered ranking **flips**. So "engineered features do not help" is not a safe conclusion. The cap of 125 must be justified from training data, not chosen on the test set.

Limitations of Phase 1

- **CV is scored differently from the test:** CV scores every training cycle (RUL up to 361), the test only the last cycle (RUL at most 145). This is why CV RMSE (36 to 42) exceeds test RMSE (26 to 31), and it means tuning optimizes a different objective.
- **Small grids:** two values per hyperparameter, several winners on a grid edge, XGBoost learning rate not tuned.
- **No uncertainty:** 3 folds and 100 test engines, with no confidence intervals or paired tests.
- **One subset:** only FD001 (one condition, one fault) is evaluated; regime normalization has not yet been tested on FD002/FD004.
- **No feature ablations yet:** we do not know which feature groups help, or how much `cycle` alone explains.

Next steps

Close out Phase 1

1. Run the capped-target ablation for all six models and commit it.
2. Score CV on last-cycle-like samples to match the test.
3. Widen grids; tune the XGBoost learning rate.
4. Feature ablations: drop `cycle`, add a long window.
5. Bootstrap confidence intervals over test engines.

Phase 2: sequence models

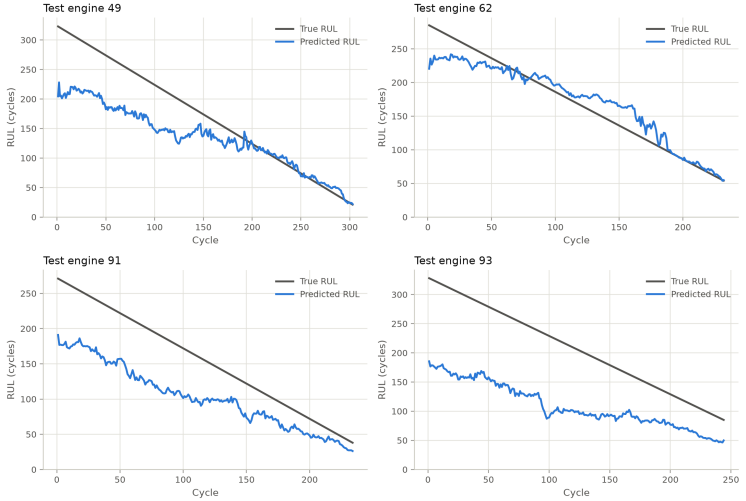
- LSTM / temporal CNN / Transformer on fixed-length sensor windows built inside engine boundaries.
- Same splits, same metrics, same last-cycle scoring.
- **Bar to beat:** 26.15 RMSE overall and about 6 to 8 RMSE in the late stage (uncapped).

Summary

- A leakage-free, engine-grouped baseline exists for FD001, with six models scored on the official test engines.
- Best uncapped test RMSE: **26.15** (raw Random Forest); best late-stage RMSE: **5.6** (engineered XGBoost).
- Engineered features clearly help near failure and hurt on young engines; overall rankings are within noise.
- The raw-vs-engineered conclusion depends on how RUL is defined, so Phase 1 is not finished until the capped ablation is run properly.

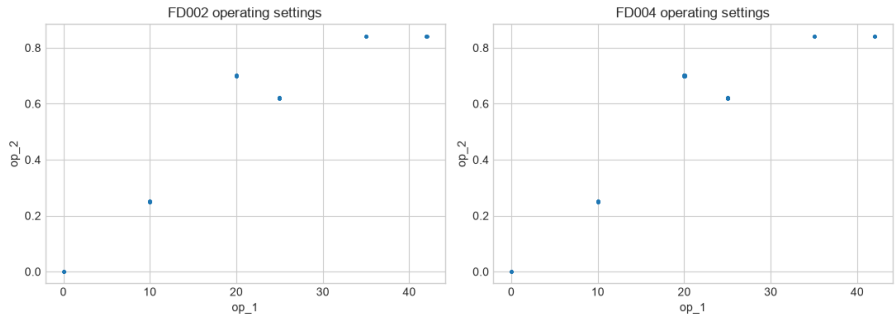
Questions?

Backup: prediction trajectories



The four test engines with the longest recorded histories, engineered model with the lowest test RMSE.

Backup: operating settings in FD002 and FD004



Six tight clusters, one per simulated flight condition. This is what the K-means regime step recovers.